MIL-S-13565D(ME)
17 May 1977
SUPERSEDING
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1 April 1971

MILITARY SPECIFICATION

SINKS, PHOTOGRAPHIC PROCESSING, TEMPERATURE-CONTROLLED

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers photographic processing sinks.

1.2 Classification. The sinks shall be of the following types and sizes, as specified (see 6.2):

Type I - For three 9- by 14-inch, outside dimension, straight-sided trays.
Type II - With refrigerated storage cabinet.
Size 1 - For three 20- by 24-inch capacity trays.
Size 2 - For three 30- by 40-inch capacity trays.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 6740

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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conditions stated for type I and type II ASIS

SPECIFICATIONS

FEDERAL

L-P-383	- Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated.
BB-F-1421	- Fluorocarbon Refrigerants.
HH-I-573	- Insulation, Thermal, (Flexible Unicellular Sheet and Pipe Covering).
QQ-S-781	- Strapping, Steel, and Seals.
QQ-W-423	- Wire, Steel, Corrosion-Resisting.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-T-60	- Tape: Packaging, Waterproof.
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-26514	- Polyurethane Foam, Rigid or Flexible, for Packaging.
MIL-A-52194	- Adhesive, Epoxy (for Bonding Glass Reinforced Polyester).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
A 249 - Welded Austenitic Steel Boiler, Superheater, Heat-Exchanger, and Condenser Tubes.
A 269 - Seamless and Welded Austenitic Stainless Steel Tubing for General Service.
A 296 - Corrosion-Resistant Iron-Chromium, Iron-Chromium-Nickel, and Nickel Base Alloy Castings for General Application.
A 312 - Seamless and Welded Austenitic Stainless Steel Pipe.
A 366 - Steel, Carbon, Cold-Rolled Sheet, Commercial Quality.
A 380 - Cleaning and Descaling Stainless Steel Parts, Equipment, and Systems.
D 1785 - Poly (Vinyl Chloride) (PVC) and Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe, Schedules 40, 80, and 120.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A5.4 - Corrosion-Resisting Chromium and Chromium-Nickel Steel Covered Welding Electrodes.
A5.9 - Corrosion-Resisting Chromium and Chromium-Nickel Steel Welding Rods and Bare Electrodes.

(Application for copies should be addressed to the American Welding Society, 2501 North West Seventh Street, Miami, FL 33125.)

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COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 5690 - Steel Wire, Corrosion and Heat Resistant.
18.5Cr-13Ni-2.5Mo(SAE 30316).

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The sinks shall consist of a base-mounted insulated trough, a water-circulation system, and a temperature-control unit (see Figures 1 and 2 for Type II sinks).

3.2 First article (first-produced sink). The contractor shall furnish one sink for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce sinks that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9).

3.3.1 Steel.

3.3.1.1 Corrosion-resisting steel plate, sheet, or strip. Corrosion-resisting steel plate, sheet, or strip shall conform to ASTM A 167, Type 316, Finish No. 2D or 4, as specified herein; castings, if any, shall conform to ASTM A 296, Class CF8M.

3.3.1.2 Corrosion-resisting steel wire. Corrosion-resisting steel wire shall conform to QQ-W-423, form optional, Composition 316, Condition A, or SAE AMS 5690E.

3.3.1.3 Corrosion-resisting steel tubing. Corrosion-resisting steel tubing shall conform to ASTM A 249 or A 269, Grade TP 316.

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3.3.1.4 Corrosion-resisting steel pipe. Corrosion-resisting steel pipe shall conform to ASTM A 312, Grade TP 316.

3.3.1.5 Carbon steel, cold-rolled. Cold-rolled carbon steel shall conform to ASTM A 366.

3.3.2 Glass-reinforced laminates. Glass-reinforced laminates shall conform to L-P-383, Type I.

3.3.3 Pipe. Polyvinyl chloride pipe shall conform to ASTM D 1785.

3.4 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain cocks, valves, and outlets shall be located for accessibility.

3.5 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.6 Base. The base shall be welded carbon steel, shall have removable panels for access to the temperature control unit, and shall be equipped with leveling screws and locking nuts.

3.7 Insulated trough. The trough shall consist of two pans, one within the other, with insulation in between, and shall be equipped with a drain. The trough shall be supported and braced to prevent buckling and sagging when filled with water. The trough shall overhang the front and sides of the base approximately 3 inches. The pans shall be corrosion-resisting sheet steel of not less than 0.063-inch nominal thickness or glass-reinforced laminate, as specified (see 6.2). Steel surfaces exposed to the developing process shall conform to ASTM A 167, Finish 4. Steel joints shall be welded; plastic joints shall be cemented with an epoxy resin cement conforming to MIL-A-52194. The inlet and outlet for the temperature-controlled circulated water shall be provided with strainers and shall be located so as to assure maximum circulation and uniform temperature of the water throughout the trough.

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3.7.1 Inner pan. The bottom of the inner pan shall have rounded upward protrusions not less than 1/4 inch high, approximately 1-3/4 inches in diameter, and not more than 8 inches center-to-center. The protrusions shall show no stretching or fatigue. The inside pan shall not leak.

3.7.2 Insulation. The walls and bottom of the trough shall be insulated with not less than 1-1/2 inches of polyurethane foam conforming to MIL-P-26514, Type I, Class 1, or Type II, Class 2, grade and category optional. Insulation shall be verminproof.

3.7.3 Dump gutter. A dump gutter or trough, when provided, shall not intrude on the inside dimensions of the sink and shall not drain into the water-circulating system.

3.7.4 Overflow outlet. An overflow outlet, draining into the drain specified in 3.7, shall be provided which will prevent the circulating water from reaching a depth greater than 2-1/2 inches at the lowest point of the inner pan.

3.8 Water-circulating system. The water-circulating system shall consist of a pump and all necessary plumbing.

3.8.1 Pump. The pump shall be corrosion-resistant and shall have a mechanical rotary seal. If located at other than the lowest point in the system (except for the drain), the pump shall be of the integral self-priming type.

3.8.2 Plumbing. All joints shall be secure against leakage and the entry of joint-sealing compounds into the system. Except where temperature dissipation is required, piping shall be insulated with insulation conforming to HH-I-573, class as required.

3.8.2.1 Water. All pipes and fittings coming in contact with the circulating water shall be corrosion-resisting steel or polyvinyl chloride plastic. Where necessary for servicing or for the replacement of components, joints in the water lines shall be made separable by the insertion of unions. Evaporative coils in contact with the circulating water shall be corrosion-resisting steel tubing and shall be one continuous length.

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3.8.2.2 Refrigeration. Plumbing in the refrigeration unit shall be copper. Fittings used in the refrigeration system shall be seamless and of the soldered type; all joints shall be silver-soldered.

3.9 Temperature-control unit. The temperature-control unit shall consist of a refrigeration unit, an electric water-heating element, and the necessary thermostatic controls to regulate the temperature of the circulating water. After the liquid in the trays has reached 68° F, the automatic controls shall maintain this temperature within 1° F throughout the ambient temperature range between 45° F and 110° F.

3.9.1 Refrigeration unit. The refrigeration unit shall consist of an electric-motor-driven, hermetically-sealed-type compressor; thermostatically-controlled solenoid valves; a condenser; sight glass; disposable drier-filter; and refrigerant expansion coils with an adjustable-type expansion valve. The unit shall be charged with dichlorodifluoromethane gas conforming to BB-F-1421, Type 12. The unit shall incorporate pressure gages for both high and low sides.

3.9.1.1 Condenser. Unless otherwise specified (see 6.2), the condenser of the refrigeration unit shall be air-cooled. The cooling air shall be circulated by a fan. The exhaust outlet shall be designed so that ducts may be readily attached.

3.9.2 Water-heating element. The water-heating element shall be encased in corrosion-resisting steel conforming to ASTM A 167, Finish 2D, and shall be of the electric-immersion type. The heater shall heat the liquid in the trays, by conduction, from 45° F to 68° F, plus or minus 1° F, in not more than 1 hour.

3.10 Electrical requirements. The sinks shall operate on a 120-volt, 60-Hertz, single-phase power source, or a 208-volt, 60-Hertz, 3-phase, 4-wire power source, as specified (see 6.2). Motors shall conform to NEMA MG1 and shall be provided with thermal magnetic inverse time overload circuit breakers. Wiring shall conform to NFPA No. 70. All electrical components and accessories shall be wired to a terminal box containing the terminals for the powerline lead-in. The pump motor shall have sealed ball bearings.

3.11 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by

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an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.11.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere, with the adherence or function of the varnish.

3.11.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.12 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.12.1 Lubricants. The sinks shall operate as specified herein when lubricated with military lubricants (see 6.6).

3.12.2 Delivery lubrication service. When delivered, all components except those for which preservatives are specified in Section 5 shall be lubricated where applicable.

3.13 Identification marking. The sink shall be identified in accordance with MIL-STD-130. The marking shall be applied to the sink on plates conforming to MIL-P-514, Type I, Style 3, Composition C of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plates will be both visible and legible.

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3.14 Instruction plates. Each sink shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the sink. Instruction plates shall conform to MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plates will be both visible and legible.

3.15 Treatment and painting. The portions of the sink normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type B. Unless otherwise specified (see 6.2), color shall be at the contractor's option. Corrosion-resisting steels shall be cleaned in accordance with ASTM A 380. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.16 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.17 Type I. Floor space requirements for the sink shall not exceed 34 by 36 inches. Working height shall be 36 inches, plus or minus 2 inches. The sink may be furnished without the pump and circulation system if it otherwise meets all the temperature control requirements specified herein. The capacity of the sink shall be three 9- by 14-inch, outside dimension, straight-sided trays in any side-by-side arrangement.

3.17.1 Controls. The sink shall be provided with a two-position switch marked "ON" and "OFF".

3.17.2 Drain. The drain for the sink shall be not less than 1 inch iron pipe size (IPS).

3.17.3 Cooling capacity. The refrigeration unit for the sink shall cool the water in the trays from 80° F to 68° F, plus or minus 1° F, in an ambient temperature of 110° F, in 1 hour or less.

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3.17.4 Transportability. The sink shall be of rigid construction and shall withstand the shock and vibration of road travel.

3.17.5 Base. The base with all components assembled thereon shall show no evidence of buckling, twisting, or warping of any member under a vertical weight of 500 pounds and a horizontal force of 100 pounds.

3.18 Type II. The sinks shall be as shown in Figures 1 and 2 (see 6.7) and as specified herein.

3.18.1 Controls. The sinks shall have a three-position switch with plate or dial marked "OFF", "CABINET", and "SINK". When the dial is set in the "CABINET" position, the storage cabinet shall function. When the dial is set in the "SINK" position, both the water-circulation system and the storage cabinet shall function.

3.18.2 Drain. The drain for each sink shall be not less than 1-1/4 inches IPS.

3.18.3 Storage cabinet. Each sink shall have a refrigerated storage cabinet located in the base of the sink. The storage cabinet shall be cooled to and maintained at a temperature of 68° F, plus or minus 1° F, by the refrigeration unit that cools the circulating water. Cooling of the cabinet shall be automatic and independent of the other components of the sink and shall be controlled by a separate thermostat, solenoid valve, and adjustable-type expansion valve. The cabinet shall be insulated on all six sides with a nonsettling, verminproof insulation. The door shall open to the full height and width of the cabinet and shall have a synthetic rubber or plastic seal. The interior of the cabinet shall be of corrosion-resisting steel conforming to ASTM A 167, Finish 2D, or of glass-reinforced laminate.

3.18.3.1 Rack. The cabinet shall contain a corrosion-resisting wire rack designed to hold 1-gallon solution bottles 5-21/32 inches in diameter by 13-9/16 inches in height (see 3.18.5.2 and 3.18.6.2). The rack shall slide out for access to the bottles and shall be designed for easy insertion and removal of the bottles.

3.18.3.2 Drip pan. The cabinet shall be provided with a drip pan made of corrosion-resisting steel conforming to ASTM A 167, Finish No. 2D. The drip pan shall fit under the coils to catch condensation and shall be removable without tipping.

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3.18.4 Base. The base with all components assembled thereon shall show no evidence of buckling, twisting, or warping of any member under a vertical weight of 1,000 pounds and a horizontal force of 100 pounds.

3.18.5 Size 1.

3.18.5.1 Cooling capacity. The refrigeration unit shall cool water in the trays and the storage cabinet from 80° F to 68° F, plus or minus 1° F, in an ambient temperature of 110° F, in not more than 2 hours.

3.18.5.2 Rack capacity. The rack in the storage cabinet shall have a capacity of at least six bottles and shall support them near the neck and base.

3.18.5.3 Transportability. The sinks shall be of rigid construction and shall withstand the shock and vibration of road travel.

3.18.6 Size 2.

3.18.6.1 Cooling capacity. The refrigeration unit shall cool water in the trays and the storage cabinet from 80° F to 68° F, plus or minus 1° F, in an ambient temperature of 110° F, in not more than 3 hours.

3.18.6.2 Rack capacity. The rack or racks in the storage cabinet shall have a capacity of at least eight bottles and shall support them near the base.

3.19 Workmanship.

3.19.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.19.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welded joints shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. Welding on corrosion-resisting steel shall be accomplished by inert gas, atomic hydrogen, or shielded-arc method. The welds shall be ground flush when practicable, polished, and passivated. Welding rods and bare electrodes shall conform to AWS A5.9. Shielded-metal-arc electrodes shall conform to AWS A5.4. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced sink inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced sink inspection.

4.3.1 Examination. The first-produced sink shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.3.2 Tests. The first-produced sink shall be tested as specified in 4.5.2.1 through 4.5.2.3.2, 4.5.2.4 when specified (see 6.2), and 4.5.2.7. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each sink shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each sink shall be tested as specified in 4.5.2.1, except the time for the test shall be 1/2 hour, and 4.5.2.5 through 4.5.2.7. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The sinks shall be examined as specified herein for the following defects:

Major

101. Parts or components not as specified.
102. Materials not as specified.
103. Safety devices not as specified.
104. Lubrication devices not as specified.
105. Drainage not as specified.
106. Visible leaks in the water system.
107. Dimensions not as specified.
108. Treatment and painting not as specified.
109. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.
202. Instructions missing or not as specified.

4.5.2 Tests. Prior to test, the sinks shall be lubricated with the oils and greases specified in 3.12.1.

4.5.2.1 Inner pan leakage. Before the trough is assembled, fill the inner pan to within 1/2 inch of the upper edge with water to which approximately 3 ounces of potassium permanganate crystals have been added for color. Wipe all exterior surfaces of the sink clean and dry, and place white paper on the floor to cover the projected area

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of the sink plus an additional 6 inches on all sides. After the solution has been allowed to remain in the pan for not less than 8 hours, examine the outside of the pan and the paper under the sinks. Leaks, as evidenced by color, shall constitute failure of this test.

4.5.2.2 Base strength. Load the completed sink uniformly with a dead weight of 500 pounds for Type I and 1,000 pounds for Type II. Apply a horizontal force of 100 pounds at the top of the base perpendicular to each of the four sides, in turn. Evidence of buckling, twisting, or warping shall constitute failure of this test.

4.5.2.3 Temperature-control test conditions. Fill the sink to operating capacity with water, and on Type II sink place the appropriate number of bottles filled with water in the cabinet. Using trays for the type of sink being tested, place the trays in the sink and fill with liquid to a depth of not less than 1-1/2 inches. Place a thermometer in each tray and, on Type II sinks, place a recording or a maximum-minimum thermometer near the center of the inside of the storage cabinet.

4.5.2.3.1 Cooling. Place the sink in an ambient temperature of 110° F, allow to stabilize, and maintain this temperature for 6 hours. Fill the sink with water and the trays with liquid, both at a temperature of 80° F. Turn on the sink and record the time required for the temperature(s) to reach 68° F, plus or minus 1° F. Times in excess of 1 hour for Type I sink; 2 hours for Type II, Size 1 sink; or 3 hours for Type II, Size 2 sink shall constitute failure of this test. Inability to maintain a temperature of 68° F, plus or minus 1° F, in the trays or in the storage cabinet, if any, during the next 6 hours shall constitute failure of this test.

4.5.2.3.2 Heating. Place the sink in an ambient temperature of 45° F or below, allow to stabilize, and maintain this temperature for 6 hours. Fill the sink with water and the trays with liquid, both at a temperature of 45° F. Turn on the sink and record the time required for the temperature(s) to reach 68° F, plus or minus 1° F. Time in excess of 1 hour shall constitute failure of this test. Inability to maintain a temperature of 68° F, plus or minus 1° F, in the trays or in the storage cabinet, if any, during the next 6 hours shall constitute failure of this test.

4.5.2.4 Road. When specified (see 6.2), mount the sink securely on a truck and drive the truck not less than 20 miles. Not less than 10 miles shall be over unsurfaced, stabilized, or partially surfaced earth or gravel roads. The route shall be subject to the approval of the contracting officer. The test shall include 10 abrupt starts and

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10 abrupt stops. Make five sharp turns at maximum safe speeds to the left and five to the right. At the completion of the run, operate the sink and examine. Any evidence of damage or malfunction shall constitute failure of this test.

4.5.2.5 Heating unit. Place the sink in an ambient temperature of not more than 65° F. Trays and thermometers shall be placed as in 4.5.2.3. Fill the sink with water and the trays with liquid, both at a temperature of 55° F, plus or minus 5° F, and turn on the sink. Continue to operate the sink for not less than 4 hours after the liquid in the trays has reached 68° F. Failure to maintain temperatures of 68° F, plus or minus 1° F, in the trays, or in the storage cabinet of Type II, shall constitute failure of this test.

4.5.2.6 Cooling unit. Place the sink in an ambient temperature of not less than 75° F. Trays and thermometers shall be placed as in 4.5.2.3, including the storage cabinet in Type II. Fill the sink with water and the trays with liquid, both at a temperature of 80° F, plus or minus 5° F, and turn on the sink. Continue to operate the sink for not less than 4 hours after the liquid in the trays has reached 68° F. Failure to maintain temperatures of 68° F, plus or minus 1° F, in the trays, or in the storage cabinet for Type II, shall constitute failure of this test.

4.5.2.7 Leak. After all other tests have been completed, the following two tests shall be made while the sink is operating in the cold cycle.

4.5.2.7.1 Refrigerant leak. All refrigerant connections and components shall be tested for leaks with a halide leak detector. Any evidence of leakage shall constitute failure of this test.

4.5.2.7.2 Water leak. All circulating-water components and connections shall be examined visually or tested with absorbent paper Any evidence of leakage shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

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4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

110.	Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
111.	Sink not secured to container base as specified.
112.	Water-circulating system not drained and valves not left open as specified.
113.	Repair parts, tools, and date items not preserved and secured as specified.
114.	Cabinet doors, racks, and pans not secured as specified.
115.	Air inlet and exhaust openings not covered as specified.
116.	Sink not covered with a full shroud as specified.
117.	Strapping not as specified.
118.	Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.8). The preproduction pack may be prepared utilizing either the first article or a production sink. When the first article is utilized, any packaging shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first article and production sinks.

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5.2 Preservation. Preservation shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Each sink shall be preserved in accordance with MIL-P-116, Method IIa. The sink shall be secured by bolting directly through the barrier material to the base of the shipping container specified in 5.3. Before sealing the barrier, the water-circulating system, including the pump, shall be completely drained and the drain valves left open. Repair parts and tools shall be preserved in accordance with the criteria and applicable methods of preservation in MIL-P-116. The data items shall be preserved in accordance with MIL-P-116, Method IC-1. The repair parts, tools, and data items shall be placed and secured in the storage compartment or sink in a manner to prevent movement or damage. Cabinet doors, racks, and pans shall be secured in place with tape conforming to PPP-T-97, Type I, Size 1/2 inch.

5.2.2 Level B. Each sink shall be secured by bolting to the base of the shipping container specified in 5.3. The water-circulating system, repair parts, tools, and data items shall be preserved and the cabinet doors, racks, and pans shall be secured as specified in 5.2.1 for Level A. The air inlet and exhaust openings into the temperature control unit shall be covered with barrier material conforming to PPP-B-1055, Class C-1 or C2(a). The barrier shall be secured with tape conforming to PPP-T-60, Type IV. Each sink shall then be covered with a full shroud fabricated from barrier material conforming to PPP-B-1055, Class C-1 or C2(a). The seams of the shroud shall be secured with tape specified herein.

5.2.3 Commercial. Each sink shall be preserved to afford protection against deterioration and damage during shipment and handling from the contractor to the initial destination.

5.3 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.3.1 Level A. Each complete sink shall be packed in a box conforming to PPP-B-601, Overseas Type, style optional or PPP-B-621, Class 2, style optional. When Level A preservation is specified, there shall be a 1-inch clearance between the sink and the top, sides, and ends of the box to prevent barrier damage. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable.

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Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.3.2 Level B. Packing shall be as specified in 5.3.1 for Level A except that boxes shall be Domestic Type or Class and strapping shall be Finish A.

5.3.3 Commercial. Each complete sink shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification.

5.4 Marking.

5.4.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial packaging. Commercial packaging marking shall reflect as a minimum, the item name, national stock number, quantity, contract or purchase order number, and appropriate address markings. The marking shall be applied with a waterproof material, shall be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The sinks covered by this specification are intended for use in processing photographic materials. Type I sinks and Type II, Size 1 sinks are used in truck-mounted darkrooms; Type II, Sizes 1 and 2 sinks are used in base installations. The Type II, Size 1 sink is to hold three trays for processing 20- by 24-inch sheets; the Type II, Size 2 sink is to hold three trays for processing 30- by 40-inch sheets.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2).
(c) Time frame required for submission of first-produced sink (see 3.2).
(d) Type of trough material (see 3.7).
(e) When the condenser shall be other than air cooled (see 3.9.1.1).

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(f) Whether the sinks shall operate on 120 or 208 volts (see 3.10).
(g) When color is other than the contractor's option (see 3.15).
(h) When road test is required (see 4.3.2 and 4.5.2.4).
(i) Time frame required for submission of the preproduction pack (see 5.1).
(j) Degree of preservation and degree of packing required (see 5.2 and 5.3).
(k) When other than Finish B strapping is required (see 5.3.1).

6.3 First-produced sink. Any changes or deviations of production sinks from the approved first-produced sink during production will be subject to the approval of the contracting officer. Approval of the first-produced sink will not relieve the contractor of his obligation to furnish sinks conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each sink.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of sinks.

6.6 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the sink covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.7 Figures. Figures 1 and 2 show the limiting dimensions for Type II sinks.

6.8 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the sinks in accordance with this specification.

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6.9 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification.

Custodian:
Army - ME

Review activity:
Army - EL

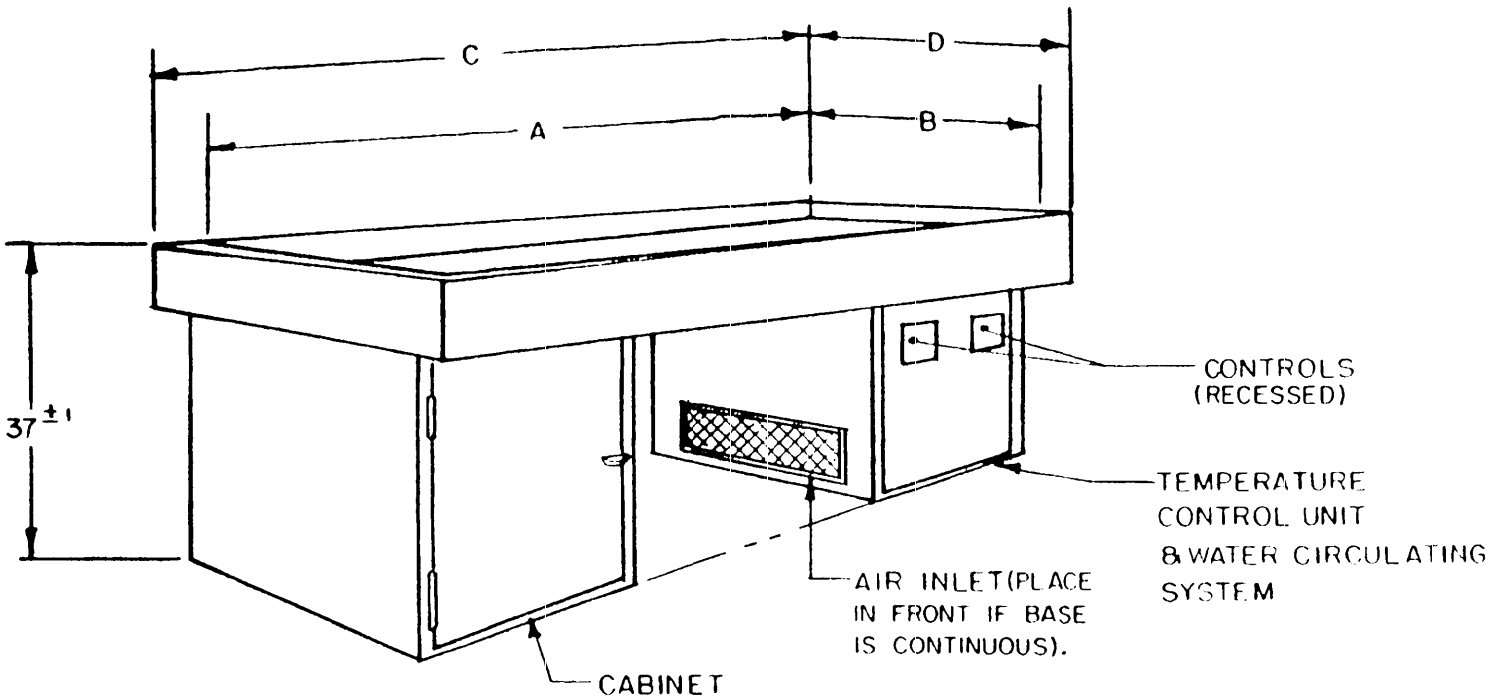
Preparing activity:
Army - ME

Project 6740-A114

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intended to have some degree of movement.

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SIZE	INSIDE TROUGH SIZE*		OVERALL DIMENSIONS	
	A	B	C	D
1	76-1/2 ± 1/2	30 + 0 − 2	80 + 0 − 1	33-1/2 ± 1-1/2
2	106 ± 2	46 ± 1	110 ± 3	50 ± 2
	* CLEAR OF INTERNAL OBSTRUCTIONS			
	INSIDE DEPTH − 5 ± 1			

NOTE:
DIMENSIONS ARE IN INCHES

FIGURE 1. Type II, temperature controlled photographic sinks.

X-2919

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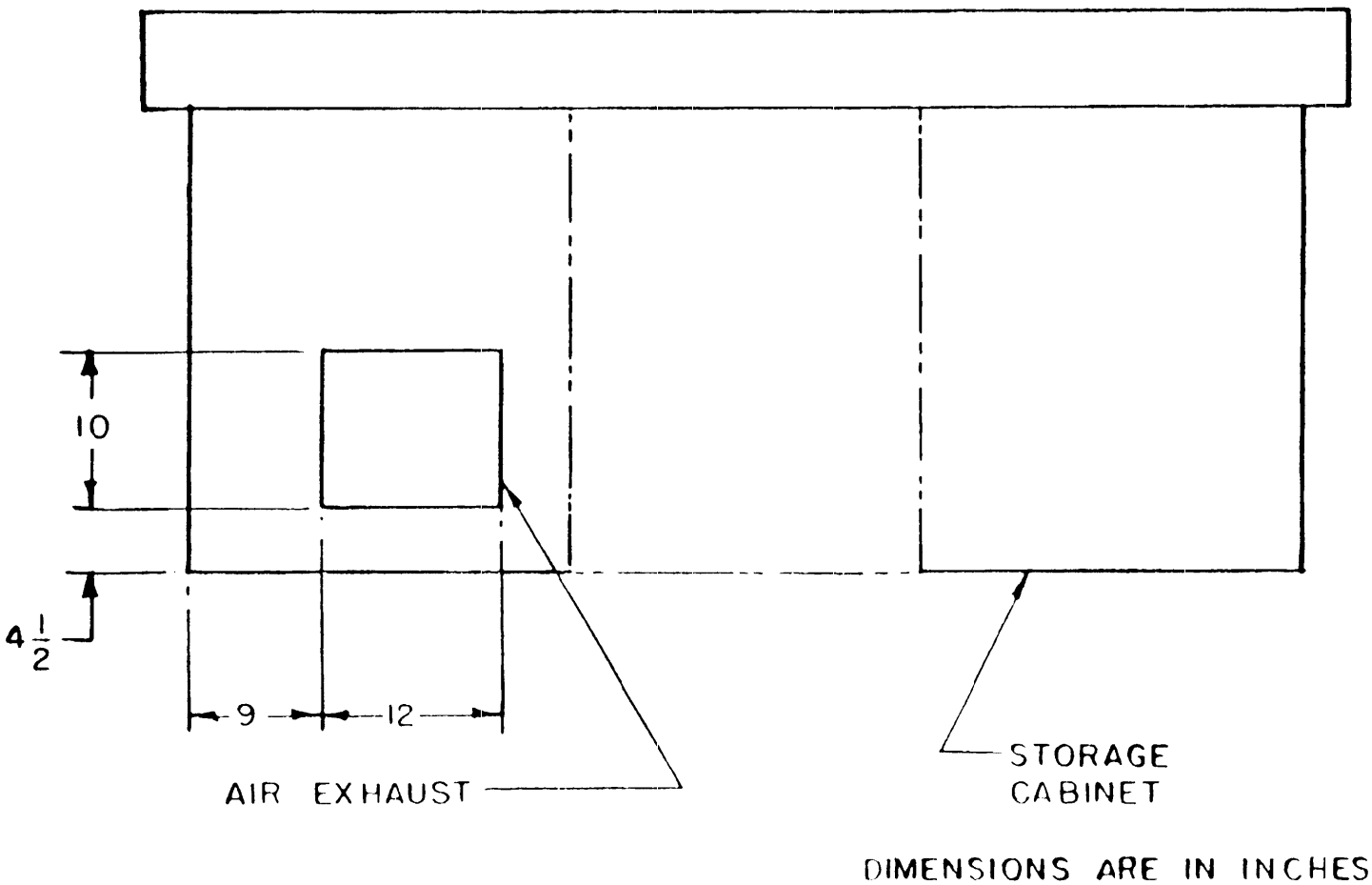


FIGURE 2. Rear view of Type II, Size I sink showing location of condenser exhaust port for truck mounting.

X-2920

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-13565D(ME) Sinks, Photographic Processing, Temperature-Controlled

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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